

Oren E. Livne
Pine Birch Analytics
35 Kelinger Rd, Churchville, PA 18966-1033
oren.livne@gmail.com

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Editor-in-Chief
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Dear Editor,

We submit for consideration the manuscript “**NLF: A Resistor-Network Framework and Linear-Time Solver for Convex Network-Flow Equilibria**” by Oren E. Livne, for publication in *SIAM Journal on Scientific Computing* in the section **Numerical Algorithms for Scientific Computing**.

The paper presents NLF, a unified solver for a broad class of convex network-flow problems — congestion (traffic) routing, minimum-delay communication routing, and exact maximum flow — cast as a single nonlinear graph Laplacian $B\rho(B^\top\phi) = \alpha d$. The solver combines a damped chord-Newton iteration on a frozen linearization with pseudo-arclength continuation through the max-flow fold, delegating each linear step to a near-linear Laplacian solver (approximate Cholesky default; LAMG+ interchangeable). The whole nonlinear solve costs 2–4 linear Laplacian solves, making the wall-clock empirically $\mathcal{O}(m)$ in the edge count across a 2,003-graph SuiteSparse corpus spanning five size decades (up to 5.6×10^7 edges). Against a state-of-the-art interior-point method, NLF is a median $2.6\times$ faster where both converge and $> 45\times$ on poorly-separable graphs where the IPM’s direct factorization is superlinear; against matrix-free L-BFGS, a median $4.2\times$ faster and the only solver to complete within budget on the 90 hardest instances.

We believe this work is well matched to SISC’s scope: it introduces a practical, reproducible method with empirical near-linear complexity, a large-scale corpus evaluation, and a unifying framework across three classical problem classes. Code, scripts, and benchmark data are openly available at <https://github.com/orenlivne/nlf>. We request the **SISC Reproducibility Badge: code and data available**.

This manuscript has not been submitted elsewhere and is not under review at another journal. All authors have approved submission.

Sincerely,

Oren E. Livne